

[Home](#) > [My Courses](#) > [Risk Management](#) > [M7: Climate Risk and the Vasicek Model](#) > Lesson Notes

Lesson Notes

MODULE 7 | LESSON 1

CLIMATE CHANGE AS FINANCIAL RISK

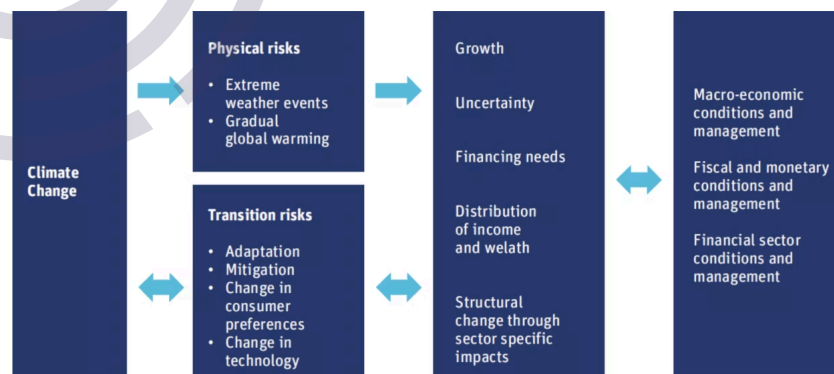
Reading Time	30 minutes
Prior Knowledge	Credit analysis, Basic Bayesian network analysis
Keywords	Transition Risk, Physical Risk, Climate Risk

In Module 5, we became very familiar with Bayesian networks, how to construct them, query them, infer from them, and sample from them. In this module, we apply what we have learned about Bayesian networks to a specific risk management problem—climate risk. We also discuss credit risk in general and how to better integrate climate risk into these types of models.

1. Climate Change as Financial Risk

You do not have to be an ecologist or an activist to appreciate that climate change affects our world, including the financial system, in very risky ways. Figure 1 below shows just some of the risk transmission vectors between climate change and the macroeconomy—and of course, the state of the macroeconomy goes hand in hand with most of the financial system and much of the rest of society.

Figure 1: Transmissions of Climate Risks to Macrofinancial Conditions



Source: Feyen, Erik et al. "Macro-Financial Aspects of Climate Change." *Policy Research Working Paper*, World Bank Group, 2020, <https://openknowledge.worldbank.org/bitstream/handle/10986/33193/Macro-Financial-Aspects-of-Climate-Change.pdf;sequence=1>.

While there is much uncertainty about the exact magnitude and timing of these impacts (because they are unprecedented), there is little uncertainty that the degree of risk we face will take many

forms. Check out this list of the “Top 10 Global Risks by Severity” according to the World Economic Forum:

1. Climate action failure
2. Extreme weather
3. Biodiversity loss
4. Social cohesion erosion
5. Livelihood crises
6. Infectious diseases
7. Human environmental damage
8. Natural resource crises
9. Debt crises
10. Geoeconomic confrontation

(Adapted from World Economic Forum 44)

We clearly recognize that most of the risks on this list are directly exacerbated by the lack of sustainable stewardship of our global natural resources, which results in detrimental climate change. The other risks are surely worsened indirectly by the same cause.

For these reasons and many more, we focus this module on modeling climate risk (with Bayesian networks) and incorporating climate risk into our credit risk models.

What is Sustainable Finance?

In the first module of this course, we discussed the many ways that the financial system is connected, or networked, and how the many connections provide pathways for financial contagion, which can snowball into a financial crisis as it spreads. In Modules 5 and 6, we continued our study of networks and network theory for this very reason. You might also recall from discussions in the Financial Markets course that finance connects with the real economy and indeed with society in general in significant ways. A very short list of examples from Financial Markets might include the innovative financial vehicles to reduce the housing crisis in Sub-Saharan Africa, the effect of real estate investment companies (REITs) on rents and housing prices in Europe and the rest of the world, and securitization as a means to finance infrastructure development, especially in India. Of course, the devastation to the macroeconomy and household wealth must figure in any discussion of the spillover between the financial system and the larger societal system it operates in.

So it comes as no surprise that there must also be significant interactions between the financial system and the state of local and global weather patterns, and we should seek to incorporate what we know about these interactions in our modeling efforts when relevant and feasible. Sustainable finance is the affirmative financial response to climate risk “to ensure investment capital is reallocated from high-carbon and low-carbon and climate resilient (LCR) options” (OECD 2).

To provide some context for the magnitude of the financial challenge and the state of the financial industry’s coordinating efforts to meet the challenge, a few data points are appropriate, as well as some information about the industry groups that are helping lead the charge: According to one measure, 128 asset managers managing **43trillioninassets—oftheworld’s** 100 trillion in global assets under management—have committed to support the goal of net zero climate-

worsening emissions by 2050 (Time). Stated as such, this “commitment” lacks the concrete specificity of many stronger, more credible commitments; nevertheless, it suggests investment that better accounts for sustainability. At the very least, the signatories to the aforementioned Net Zero Asset Managers initiative (NZAM) felt compelled to proclaim their support in response to their own investor clients’ demands, and these demands will likely become stronger as the evidence mounts for the need for such commitments. NZAM is but one of the alliances that make up a larger umbrella organization, that is, the Glasgow Financial Alliance for Net Zero (GFANZ).

GFANZ is “a global coalition of leading financial institutions committed to accelerating the decarbonization of the economy” that is headed by two of the most recognizable names in finance, with Mike Bloomberg representing the private side of the financial sector (CEO of Bloomberg L.P.) and Jay Carney who represents both the private sector and the regulatory side as the former Governor of the Bank of England from 2013 to 2020. GFANZ makes the following powerful statement regarding the urgency of their purpose:

“Every company, bank, insurer and investor will need to adjust their business models, develop credible plans for the transition to a low-carbon, climate resilient future and then implement those plans. Private finance can help fund private sector initiatives and turn billions committed to climate investment through public channels into trillions of total climate investment. But unlocking systemic change will require collaborative, ambitious commitments and near-term action across the entire financial system.”

The United Nations’ own Investment Leadership Programme also seeks “to accelerate the pace of collaboration amongst leading investors and boost action on achieving key global sustainability objectives such as the Paris Climate Agreement and the Sustainable Development Goals (UN Investment). Still another group, The Network of Central Banks and Supervisors for Greening the Financial System (NGFS), also seeks “to enhance the role of the financial system to manage risks and to mobilize capital for green and low-carbon investments in the broader context of environmentally sustainable development” (NGFS Purpose).

All of this is just to say that sustainable investment is a growing imperative with which we must likely reckon as financial engineers, in one form or another.

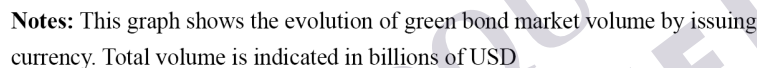
2. Green Bonds

As of 2015, green bond issuance stood at 40 billion USD, less than a fifth of the 2.26 trillion USD in needed “low-carbon investment,” which in turn composes a small fraction of the 19 trillion USD issuance by China, Japan, the United States, and the European Union in 2014. The ratio of actual to required sustainable investment is indeed small and by itself disheartening, but has nevertheless been growing more or less geometrically over the past decade, as you can see in figure 2 below.

What is a green bond? A green bond is just a bond committed “to finance or re-finance green projects, assets or business activities” (OECD 5). Unfortunately, “green” has yet to be conclusively defined and may not be for some time, inviting confusion, uncertainty, and even fraud. We will discuss this problem more in the next section.

Let us illustrate with some examples. In 2019, the Nigerian electricity supplier North South Power issued the first and second green bonds (from a non-financial company) in Sub-Saharan Africa for a total issuance equivalent to 26 million USD to finance its solar energy development. In 2021, Côte

Figure 2: Growing Green Bond Volume



Source: Cheong, Chiyong et al. "Green Bonds: A Survey." Emerald Insight, Nov/Dec 2020, <https://www.emerald.com/insight/content/doi/10.1108/JDQS-09-2020-0024/full/html>

According to Foster, greenwashing “refers to providing misleading information suggesting that products or investments are more environmentally sound, or ‘green,’ than they actually are.” That is an apt definition, but it also highlights that greenwashing depends on what *exactly* is meant by green, and the definitions for green and sustainable are many, possibly at least as many as the number of new “sustainable funds” launched in 2021: over 120. According to Foster, “there are five times as many sustainable funds in the U.S. as a decade ago, and three more than five years ago” as of 2022. As we have seen with factor investing, with such proliferation there is bound to be a degradation in the quality of what is considered “green” or “sustainable.” In response to valid concerns about greenwashing, the United States financial regulator, Securities and Exchange Commission, has proposed rules requiring fund managers to justify their green and sustainable labeling (Foster).

The reading by Wu et al. for this lesson mostly pertains to our course in a few ways:

- <https://learn.wgu.edu/my-path/courses/risk-management/modules/m-7-climate-risk-and-the-vasicek-model/tasks/lesson-1-clima...> 4/6

risk that would be of natural interest to many fundamental analysts in many sectors, where real estate and insurance are just the most obvious examples.

2. Wu et al. use Bayesian networks to perform their analysis, so we can observe an extensive and very practical application of this valuable tool—as well as as the steps they took to execute their research, e.g., literature review, risk factor identification, criteria selection, data collection, model specification and parameterization, performance testing and sensitivity analysis (Wu 2166–2178)—steps to follow or adapt in similar research you may choose to pursue.
3. What's more, Wu et al. combine the traditional Bayesian network with another technology (Geographic Information System, or GIS) in an extension of Bayesian network analysis that should inspire further innovation with other technologies in the similar ways or in completely novel ways. GIS functions mostly to collect the data and store the data, as well as to perform some initial analysis (2180), so you might consider how some alternative source of collected and processed data might be substituted for GIS depending on your particular future project or challenge.
4. This article discusses the “problem” of subjectivity (versus objective data) at several points. This discussion will continue in the next module, where subjectivity is rather embraced as a source of knowledge that should not be left unexploited in the scientific search for descriptive models. As Wu et al. state, the Bayesian network model lends itself to “a more scientific method that can combine quantitative and qualitative analysis to assess” risk (2164).

5. Conclusion

Now that we have read about approaches to incorporating climate risk in our credit risk models, we will look at a specific example that mirrors the Chinese flood example we read about in Wu et al., in the form of a Python notebook.

References

- Cheong, Chiyong et al. "Green Bonds: A Survey." *Emerald Insight*, Nov/Dec 2020, https://www.emerald.com/insight/proxy/img?link=resource/id/urn:emeraldgroup.com:asset:id:article:10_1108_JDQS-09-2020-0024/urn:emeraldgroup.com:asset:id:binary:JDQS-09-2020-002401.tif
- Feyen, Erik et al. "Macro-Financial Aspects of Climate Change." *Policy Research Working Paper*, World Bank Group, 2020, <https://openknowledge.worldbank.org/bitstream/handle/10986/33193/Macro-Financial-Aspects-of-Climate-Change.pdf;sequence=1>.
- Foster, Lauren. "How 'Greenwashing' Cost a CEO His Job." *Barron's*, 2 Jun 2022, <https://www.barrons.com/articles/greenwashing-dws-group-ceo-51654130739>.
- Murdoch, Adrian. "Why Côte d'Ivoire's first green bond is so important." *Capital Monitor*, 23 Aug 2021, <https://capitalmonitor.ai/sector/consumer/why-cote-divoires-first-green-bond-is-so-important/>.
- OECD, "Green bonds: Mobilising the debt capital markets for a low-carbon transition." *OECD Policy Perspectives*, Dec 2015, <https://www.oecd.org/environment/cc/Green%20bonds%20PP%20%5Bf3%5D%20%5Blr%5D.pdf>.
- World Economic Forum. *The Global Risks Report*. World Economic Forum, 2022, 17th ed., https://www3.weforum.org/docs/WEF_The_Global_Risks_Report_2022.pdf.

- Wu, Zening et al. "Assessing Urban Flood Disaster Risk using Bayesian Network Model and GIS Applications." *Geomatics, Natural Hazards and Risk*, vol. 10, no. 1, pp. 2163–2184, <https://www.tandfonline.com/doi/full/10.1080/19475705.2019.1685010>.

